

Advanced Motion Control

Part IV: Repetitive Control

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2025/2026

RC: When to Use?

Limitations of Classical Control

RC: Basic Principles

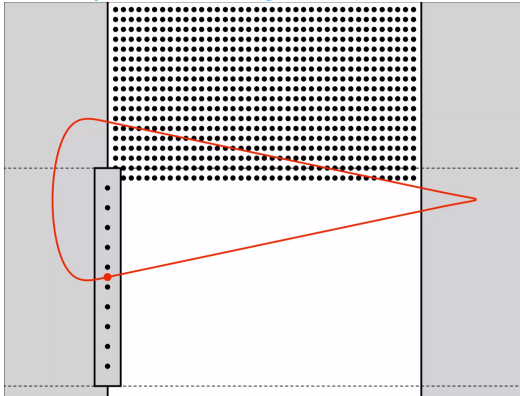
Enforcing Stability

Design & Implementation Aspects

Example

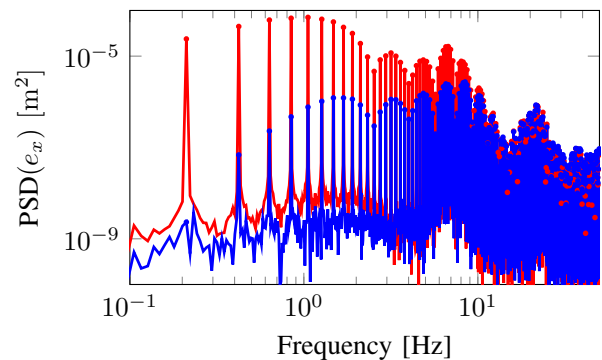
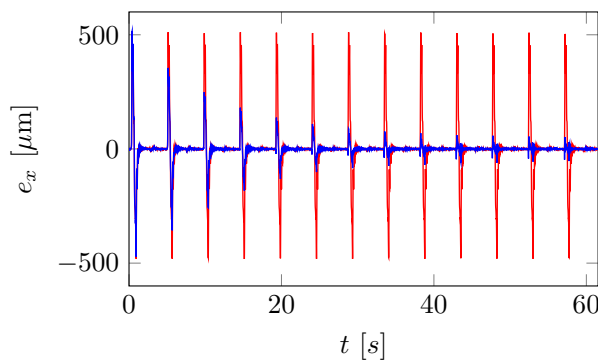
Extensions

Example: Printing with Continuous Media Flow (Blanken, Koekebakker & Oomen 2020)



Example: Printing with Continuous Media Flow

- Classical control: repeating error
- Repetitive control: 90% peak reduction



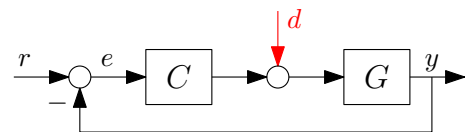
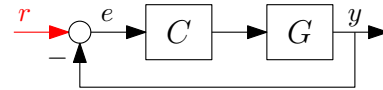
A range of applications

Periodic tasks

- ▶ Pick-and-place machines
- ▶ Wafer scanners
- ▶ Printers (printhead motion)
- ▶ ...

Periodic disturbances

- ▶ Rotating mechanisms:
 - ▶ reading optical discs (CDs, ...)
 - ▶ printers (paper handling)
 - ▶ wind turbines
- ▶ Drive control with nonuniform torque pattern
- ▶ ...



Repetitive Control: When to Use?

Example: servo problem, e.g., pick-and-place machine

General setup: ILC

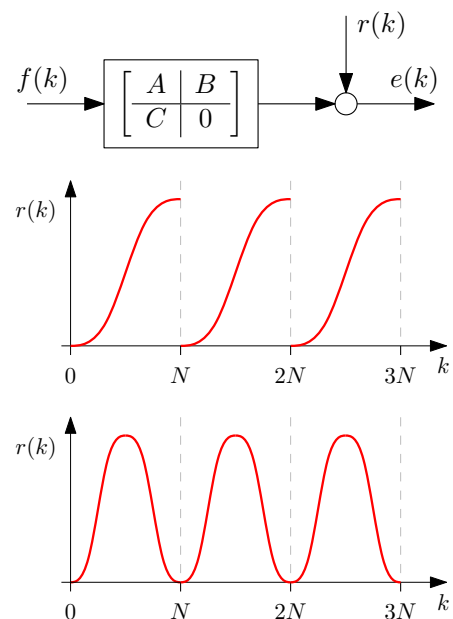
- ▶ Repeating disturbance $r(k)$
- ▶ $x(0) = 0$, and $x(0) = x(N) = x(2N) = \dots$

System operates over *independent* iterations

General setup: RC

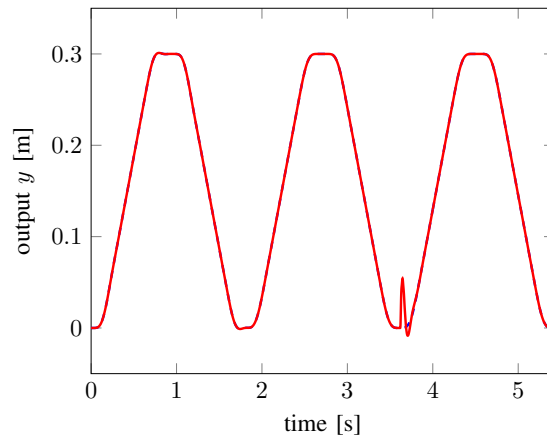
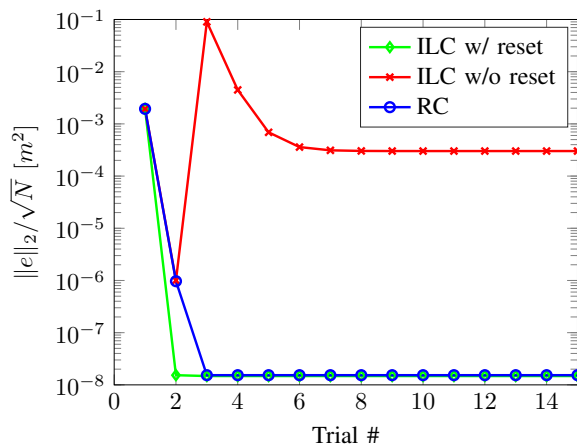
- ▶ Periodic disturbance $r(k)$
- ▶ $x(0) = 0$, but $x(N), x(2N), x(3N)$ depend on $f(k)$

System *evolves continuously* in time



What's the difference?

- ▶ Let's apply ILC... in an RC setting
- ▶ **This does not work!!** $\Rightarrow$ due to falsely assumed state reset (e.g., system is not at 'rest')



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Limitations of Classical Control

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Feedback control

Aim: deal with uncertainty

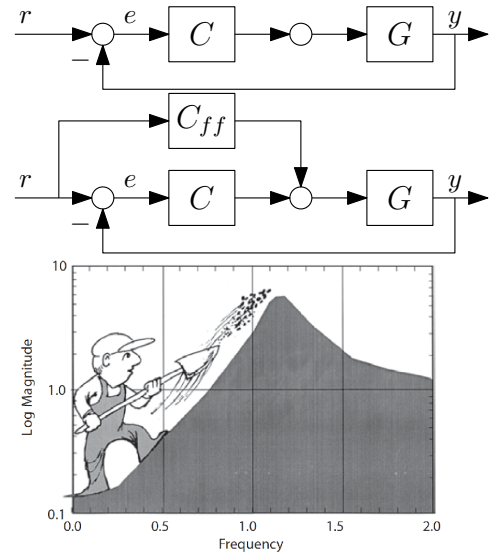
- ▶ system dynamics
- ▶ disturbances

Performance limit: causality $\Rightarrow$ Bode sensitivity integral

Feedforward control

Aim: performance

- ▶ depends on model quality, $C_{ff} \approx G^{-1}$
- ▶ non-causality is allowed (no waterbed effect)
- ▶ ... if disturbance is known beforehand



Limitations of Classical Control

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Feedback control - revisited

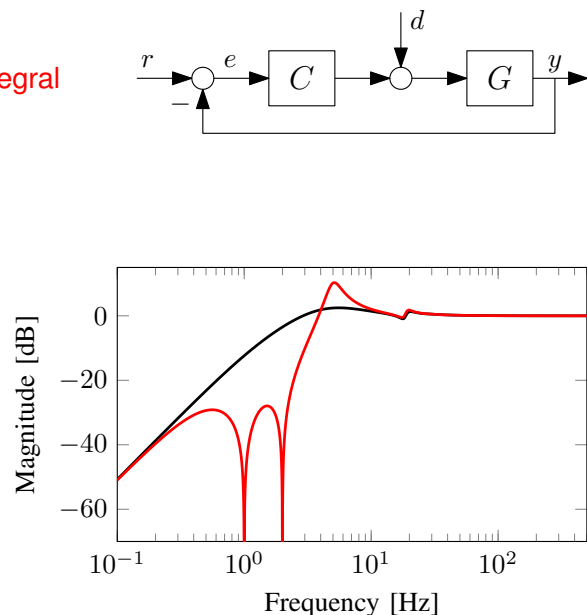
Performance limit: causality $\Rightarrow$ Bode sensitivity integral

Observation: for periodic d , $e = 0$ only requires S has zero gain at harmonic frequencies

Possible approach:

- ▶ Inverse notch filters at harmonics
- ▶ **Problems:**
 - ▶ Guaranteeing stability cumbersome task (manual loopshaping)
 - ▶ What if disturbance period changes? (Manual) redesign required?

Next: a systematic design solution (RC)



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The Idea of Repetitive Control

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- ▶ Learn from previous (error/disturbance) repetitions
- ▶ then we can **beat causality**: we 'know' the future disturbances!
- ▶ Developments in RC:
 - ▶ initial idea: 1980 paper (Inoue et al. 1980) (in Japanese) "High accuracy control magnet power supply of proton synchrotron in recurrent operation"
 - ▶ stability analysis and design: (Hara et al. 1985, 1988, Tomizuka et al. 1989)
 - ▶ 1990-present: many results, well-established theory, many applications
- ▶ Key theoretical foundation: Internal Model Principle (IMP) (Francis & Wonham 1975)

Internal Model Principle

Asymptotic rejection is achieved if a model of the disturbance generating system is included in the feedback loop (Francis & Wonham 1975)

Example:

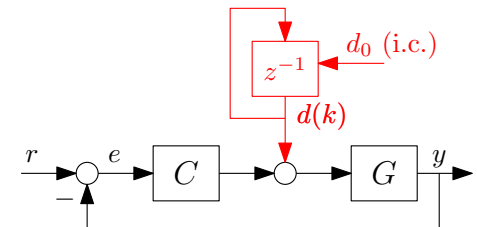
- Step disturbance $d(k) = \begin{cases} d_0, & \text{for } k \geq 0 \\ 0, & \text{for } k < 0 \end{cases}$

Take z -transform:

$$D(z) = \mathcal{Z}\{d(k)\} = \sum_{k=-\infty}^{\infty} d(k)z^{-k} = \frac{1}{1-z^{-1}} D_0$$

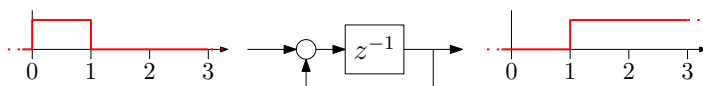
$\Rightarrow$ include **integrator** in control loop!

- Observation: $d(k)$ is periodic signal with period $N = 1$
What about arbitrary periodic signals?

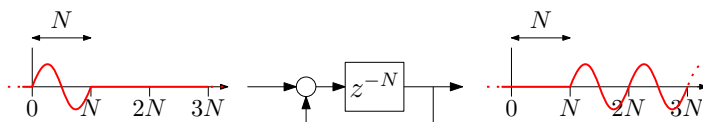


Periodic signal models

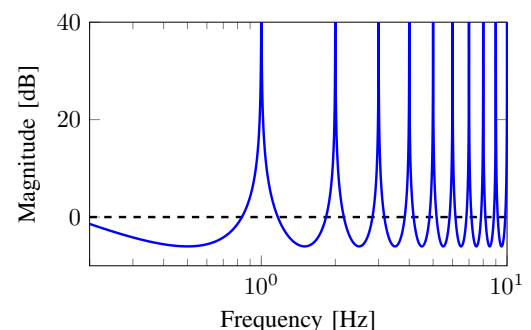
- Step disturbance: integrator



- General N -periodic disturbance: memory loop (N integrators)

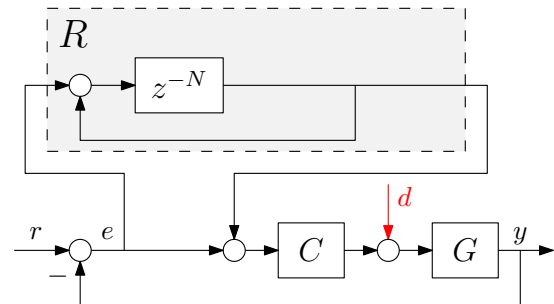
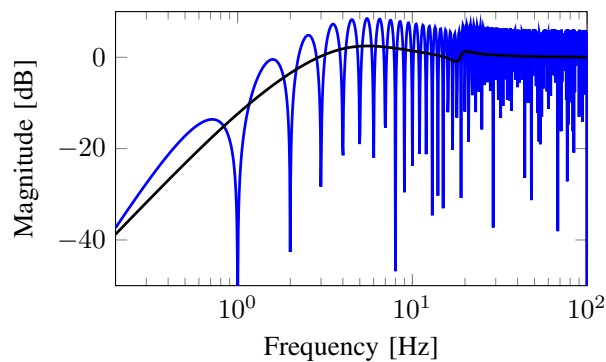


- Next: include in control loop! (IMP)



Add-on RC

- ▶ Add-on to standard feedback
- ▶ Memory loop included in control loop
- ▶ Infinite rejection at harmonic frequencies!
- ▶ What about closed-loop stability?



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Stability Analysis

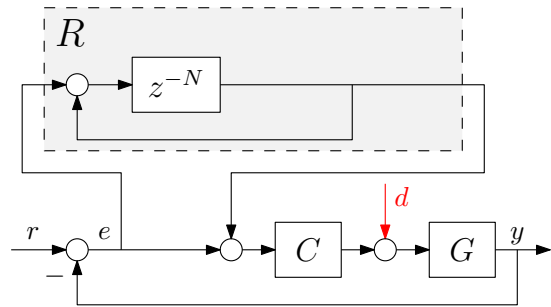
- Closed-loop transfer function:

$$\begin{aligned}
 e &= (I + GC(I + R))^{-1}(r - Gd) \\
 &= \underbrace{(I + TR)^{-1}}_{S_R} \underbrace{(I + GC)^{-1}}_S (r - Gd) \\
 &= S_R S (r - Gd)
 \end{aligned}$$

where $T = GC(I + GC)^{-1} = I - S$

- S is sensitivity 'without RC'
- S_R is called 'modifying sensitivity'
- S , SG and T are assumed stable due to design C

Key question: S_R stable?



Stability Analysis

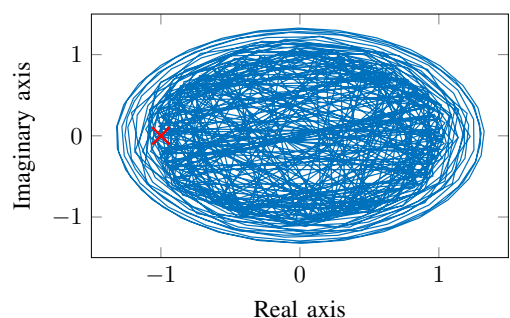
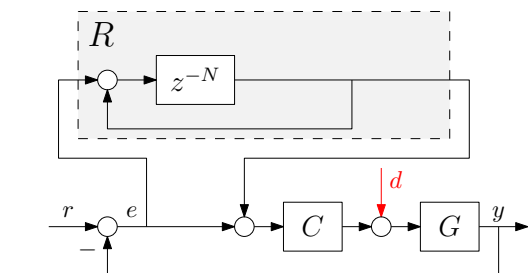
Key question: S_R stable?

$$S_R = \frac{1}{1 + TR} = \frac{1}{1 + T \left(\frac{z^{-N}}{1 - z^{-N}} \right)} = \frac{1 - z^{-N}}{1 - z^{-N}(1 - T)}$$

Nyquist stability theorem for RC

Assume S , T are stable. Then, S_R is stable if and only if the Nyquist plot of $-z^{-N}(1 - T)$

- makes no encirclements of the point -1 , and
- does not pass through the point -1
- $|1 - T| = |S| > 1$ for some frequencies
- z^{-N} gives 360° phase shift every $1/NT_s$ Hz



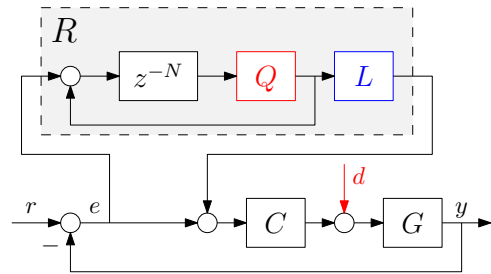
Design for Stability

$$S_R = \frac{1 - z^{-N}Q(z)}{1 - z^{-N}Q(z)(1 - T(z)L(z))}$$

Enforce stability by adding

- ▶ Learning filter L
- ▶ Robustness filter Q

Note similarities with ILC!



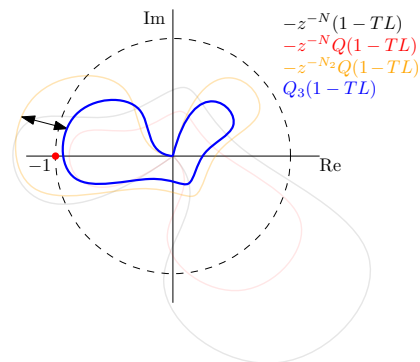
Nyquist stability theorem for RC

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- ▶ does not pass through the point -1

RC design procedure

- ▶ Design L to approximate T^{-1}
- ▶ Design Q
 - ▶ for performance: $Q(\omega) \approx 1$
 - ▶ for stability and robustness: avoid encirclements
- ▶ Very useful in practice: design independent of N
 - ▶ varying N rotates Nyquist plot (note $|z^{-N}| = 1$)
 - ▶ guaranteed stability for all N : use small-gain



Nyquist stability theorem for RC

Assume S , T are stable. Then, S_R is stable if and only if the Nyquist plot of $-z^{-N}Q(z)(1 - T(z)L(z))$

- ▶ makes no encirclements of the point -1 , and
- ▶ does not pass through the point -1

Small-gain stability condition

Assume S , T are stable. Then, S_R is stable for all N if

$$|Q(z)(1 - T(z)L(z))| < 1, \forall |z| = 1$$

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Design Aspects

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RC Filter Design

- ▶ Design L to approximate T^{-1}
 - ▶ If T nonminimum phase $\Rightarrow T^{-1}$ unstable
 - ▶ If T strictly proper $\Rightarrow T^{-1}$ improper $\Rightarrow T^{-1}$ non-causal
- ▶ Design Q such that
 - ▶ $Q(e^{i\omega}) \approx 1$ (performance, otherwise IMP violated!)
 - ▶ $|Q(1 - TL)| < 1, \forall \omega \in [0, 2\pi]$ (robustness)

Stability theorem for RC design

Assume S, T are stable. Then, S_R is stable
for all N if and only if

$$|Q(z)(1 - T(z)L(z))| < 1, \quad \forall z = e^{i\omega}, \omega \in [0, 2\pi]$$

Approaches to compute L

- ▶ ZPETC (Tomizuka 1987) (see part ILC)
- ▶ $\mathcal{H}_\infty$ -synthesis with finite preview (van Zundert & Oomen 2018)
- ▶ ...

Approaches to compute Q

- ▶ $Q \neq 1$: nonperfect tracking
 - ▶ $|Q| \neq 1$
 - ▶ $\angle Q \neq 0$
- ▶ typically non-causal *zero-phase* low-pass filter
 - ▶ e.g. $Q(z) = q[-2]z^2 + q[-1]z + q[0] + q[1]z^{-1} + q[2]z^{-2}$
 - ▶ Matlab: `fir1.m`, `filtfilt.m`

How to implement preview of L and Q ?

Implementation Aspects

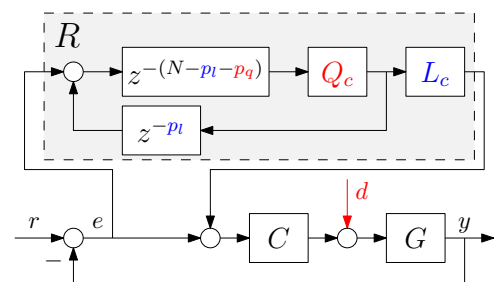
How to implement preview?

- ▶ RC is feedback (hence causal), but filters L , Q can be non-causal
- ▶ Embed preview in memory loop:
 - ▶ $L(z) = z^{p_l} L_c(z)$
 - ▶ $Q(z) = z^{p_q} Q_c(z)$

Possible for $p_l + p_q \leq N$

Key reason for exceptional performance RC

- ▶ RC anticipates on future (periods of) disturbances
- ▶ achieved through non-causal design of L , Q
- ▶ enabled by embedding in memory loop



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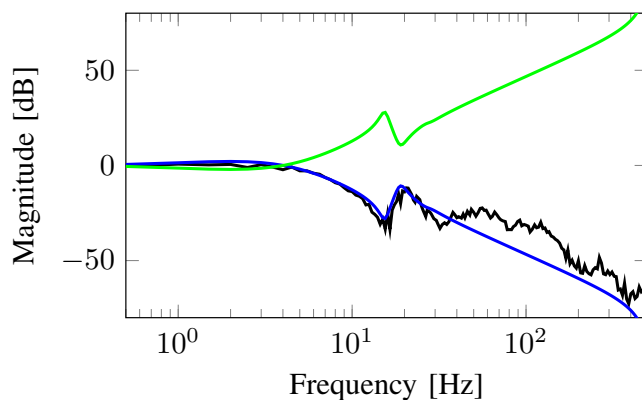
Extensions

RC: Design Example

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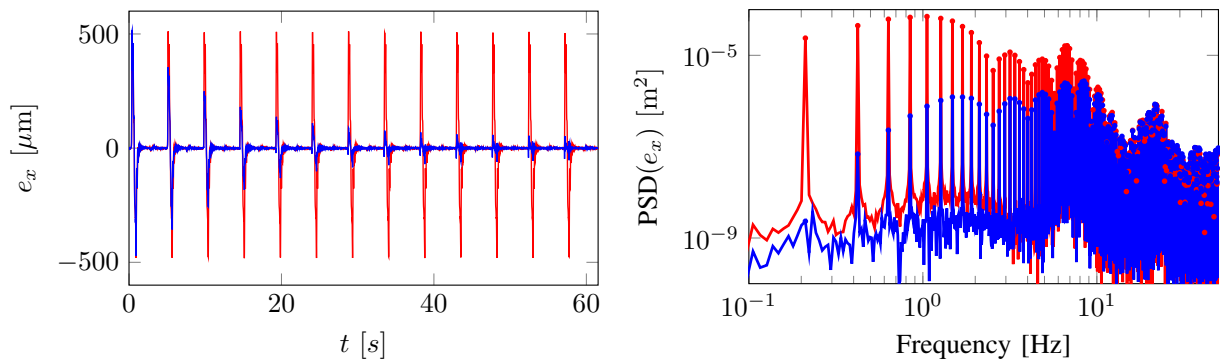
Example: Printing with Continuous Media Flow (Blanken, Koekebakker & Oomen 2020)

- Obtain model $\hat{T}$ of T
- Design L such that $L \approx \hat{T}^{-1}$, i.e., $1 - TL \approx 0$
- Design Q such that $|Q(e^{i\omega})(1 - T(e^{i\omega})L(e^{i\omega}))| < 1, \quad \forall \omega \in [0, 2\pi]$



Example: Printing with Continuous Media Flow (Blanken, Koekebakker & Oomen 2020)

- Obtain model $\hat{T}$ of T
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- Implement!



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- ▶ Higher-order RC (Steinbuch 2002, Steinbuch et al. 2007) : robustness to disturbance period variations and nonperiodic disturbances
- ▶ Multi-period RC: design for nonperiodic summations of multiple periodic disturbances (Blanken, Bevers, Koekebakker & Oomen 2020)
- ▶ Multivariable RC (Blanken, Koekebakker & Oomen 2020) : SISO decentralized design for MIMO systems
- ▶ RC for position-domain periodic disturbances: Gaussian process-based internal models (Mooren et al. 2022)

References I

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